

Loan Default Prediction

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<https://github.com/mlui2/data1030-loan-prediction-project>

Introduction:

Predicting loan defaults is a central challenge for consumer lending institutions, as misestimating borrower risk directly affects profitability, capital allocation, and borrower access to credit. In this project, I develop and evaluate machine learning models to predict whether a LendingClub loan will default prior to loan origination, using only information available at the time of the lending decision.

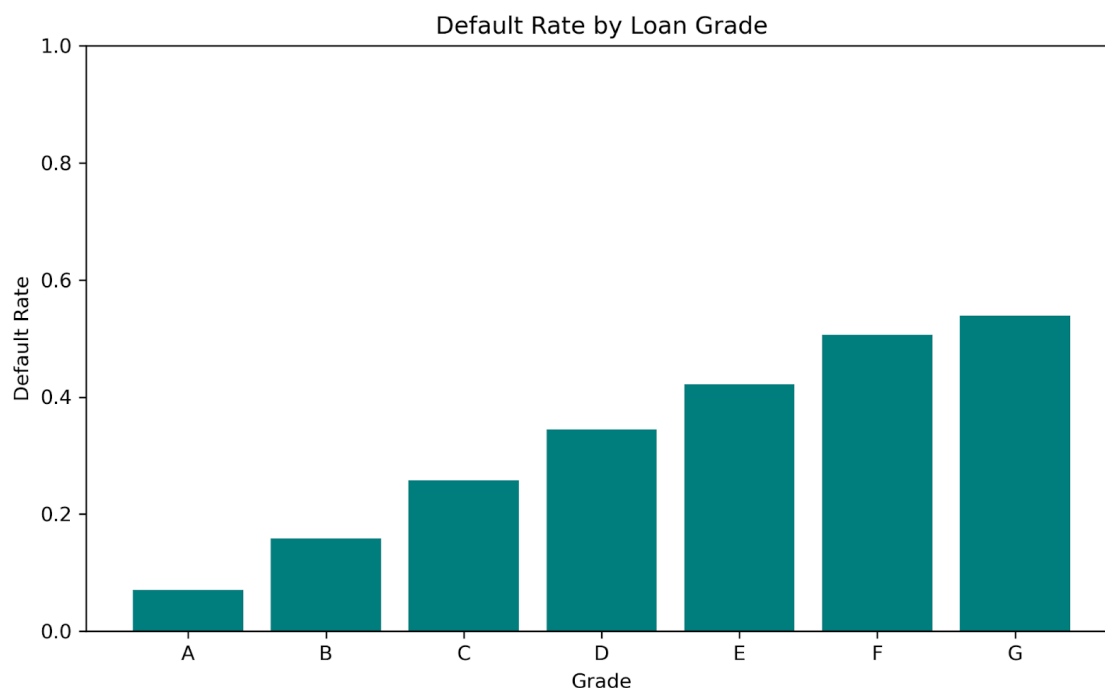
The dataset is sourced from LendingClub's public loan data spanning 2007–2018 and accessed through Kaggle. After performing data cleaning, the final dataset contains approximately 480,000 observations and 99 columns. These include 2 ordinal features, 8 categorical features, and 88 numerical features, along with a binary target variable indicating loan repayment status. The dataset is moderately imbalanced, with roughly 78% repayments and 22% defaults.

Although prior work exists on predicting defaults using the same dataset, direct comparisons across studies are difficult due to substantial methodological differences. In particular, many existing approaches include loans with unresolved repayment statuses by assigning them to either the default or repaid class, whereas this project excludes such observations entirely. In addition, this analysis removes all post-origination features to avoid information leakage, while some prior studies incorporate variables that would not be observable at loan issuance. Lastly, missingness is addressed by restricting the dataset to post 2016 loans rather than dropping features across the full dataset.

EDA:

Initial EDA suggests that default risk in our dataset may be largely structured around a small number of dominant variables. One of the clearest patterns emerges when examining default rates by loan grade, as shown in Figure 1.

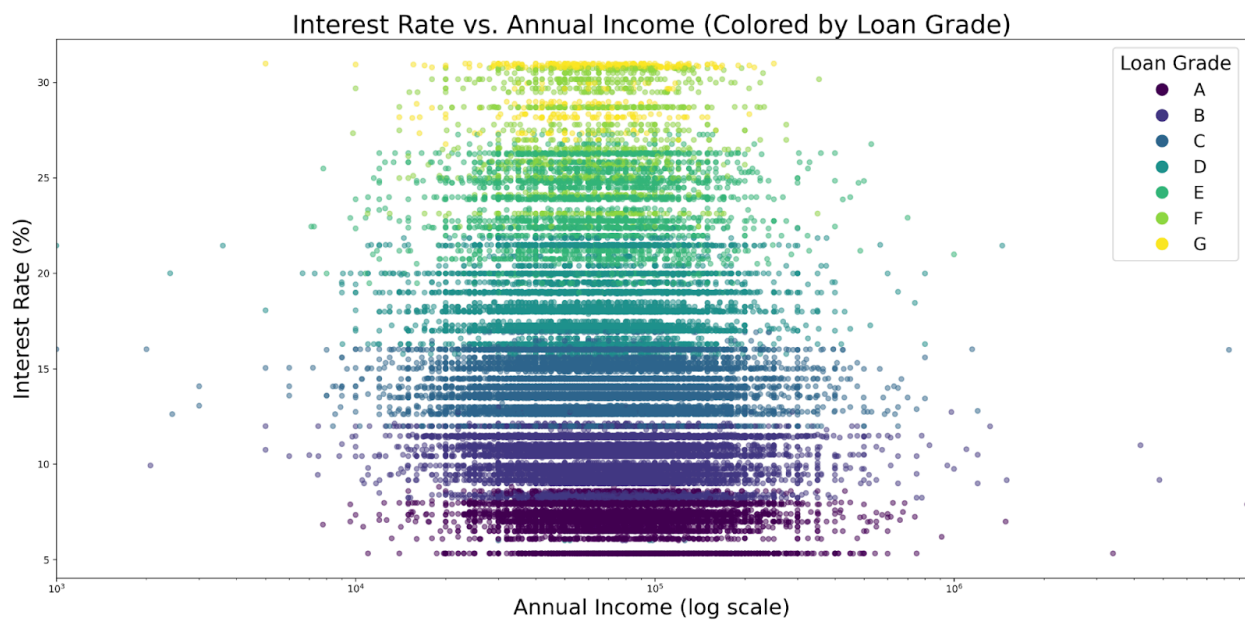
Figure 1:



Default rates increase monotonically as loan grade decreases, indicating that LendingClub's internal grading system already captures a substantial portion of borrower risk.

Additional insight is provided by examining the relationship between interest rates and annual income, colored by loan grade, as shown in Figure 2.

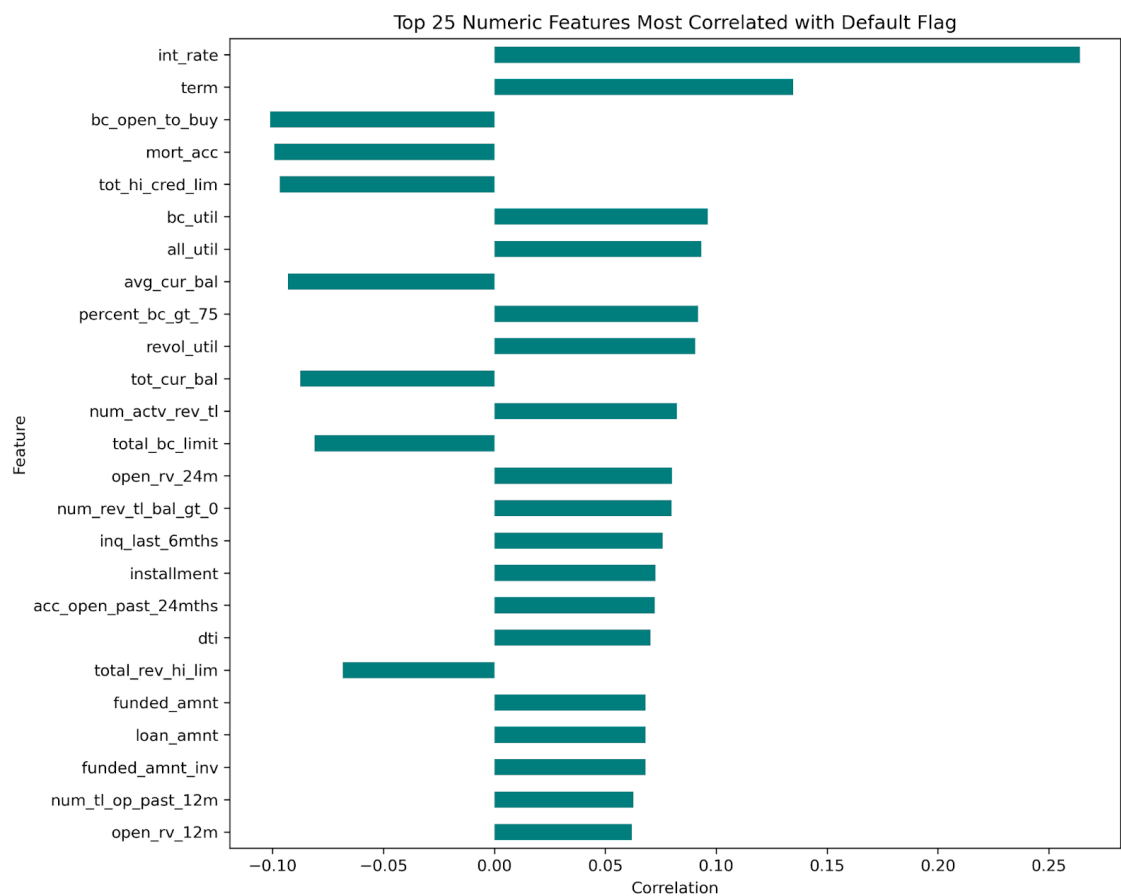
Figure 2:



Loan grades correspond closely to discrete interest rate levels, producing clear clusters and step-like jumps across grades. In contrast, annual income varies widely within each grade, with no clear relationship to loan grade. Together with Figure 1, this suggests that interest rates, which are directly linked to grade, may carry substantial information about default risk, while income alone appears less informative in isolation.

Finally, correlations between the target variable and the top 25 numeric features are shown in Figure 3.

Figure 3:



Interest rate exhibits the strongest correlation with the default indicator among numeric variables, further reinforcing its importance as a key risk-related feature. Other numeric features show considerably weaker correlations, supporting the view that a small set of dominant variables drive much of the observable structure in default outcomes.

Methods:

1. Data Splitting

To mirror realistic deployment, the dataset was split chronologically. The most recent 10% of loans, issued in 2018, were held out as a final test set, and the remaining training data underwent a 3 fold time series split. For XGBoost, an additional validation set was carved out from the training data by using 20% of the most recent observations to be used for early stopping. This ensures that all evaluations reflect forward looking predictions and prevent future information leakage.

2. Preprocessing

Categorical and ordinal features were encoded using one-hot and ordinal encoding, respectively. Missing values in categorical variables were explicitly represented using a dedicated “NA” category to preserve missingness information. Numerical features were standardized to ensure comparability across scales for models sensitive to feature magnitude.

Non-structural missing values in numerical features were handled using iterative imputation with a random forest regressor, following procedures introduced in class. Structural missingness was addressed through appropriate constant imputation combined with explicit missingness indicator variables to improve interpretability and preserve information. For XGBoost, which natively handles missing values, numerical feature imputation and scaling were omitted.

3. Evaluation Metric and Baseline

Model performance was evaluated using the F2 score, which places greater emphasis on recall than precision. This metric reflects the asymmetric costs in consumer lending, where failing to identify a borrower who will default represents a direct financial loss, while incorrectly rejecting a repaying borrower primarily corresponds to a missed opportunity. All models were evaluated using the same metric to ensure comparability.

A stratified dummy classifier was used as a baseline, predicting default outcomes according to the class distribution in the test set.

4. Models and Hyperparameters Evaluated

The algorithms and hyperparameters explored are summarized in the table below:

Model	Hyperparameters ranges
Logistic Regression	C: [0.0001, 0.001, 0.01] l1_ratio: [0, 0.2]
XGBoost	max_depth: [1, 2, 3, 5] reg_alpha: [1, 5, 8, 10, 20] reg_lambda: [500, 1000, 1500, 2000]
KNN	n_neighbors: [1, 3, 5] weights: [uniform, distance]
Decision Tree	max_depth: [1, 2, 3] min_samples_leaf: [1, 5] min_samples_split: [2, 20]
Random Forest	n_estimators: [100, 200, 300] max_depth: [1, 2, 3] max_features: [0.1, 0.2, 0.3]

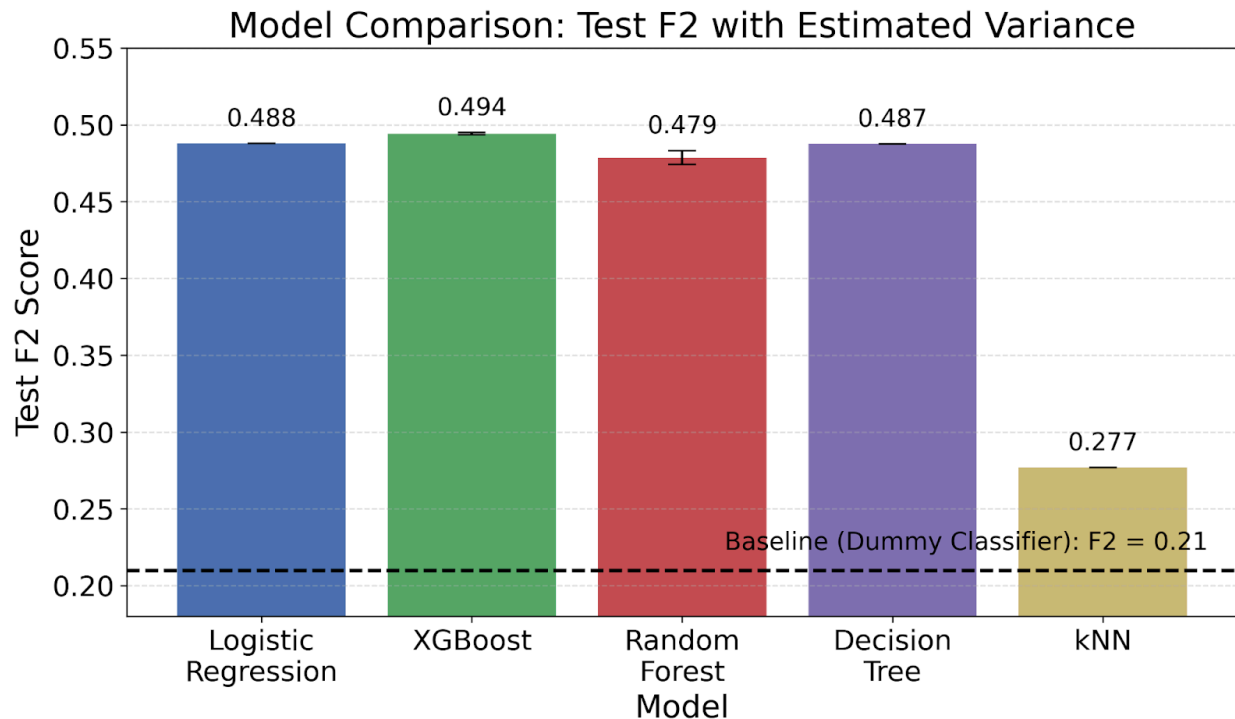
5. Uncertainty

To assess model stability and account for stochasticity in training, each model was tuned, trained and evaluated using three different random seeds. Uncertainty arising from data splitting was not measured, as introducing random splits would have disrupted the chronological structure of the dataset and allowed models to implicitly learn from future information.

Results

Model performances are summarized in figure below. For stochastic models, z scores in comparison to baseline are also shown. The best performing model was XGBoost, which achieved a mean score of 0.4942 across 3 seeds. The highest varying model was RF, with a std of 0.0046.

Figure 4:



$z(\text{RF}) \sim 58.4$

$z(\text{XG}) \sim 315.8$

1. Model Interpretations

A. Deterministic Models: (Identical parameters / scores across seeds)

1. Logistic Regression

Val	Test	Params
0.5657	0.4879	{C: 0.001}, {l1_ratio: 0}

Logistic Regression's optimal parameters correspond to pure ridge regularization with extremely strong penalization.

This choice reflects a preference for smooth coefficient shrinkage rather than feature elimination. This suggests that many features in the dataset contribute weak but useful information. Strong regularization further indicates substantial noise and multicollinearity among features.

2. Decision Tree

Val	Test	Params
0.5882	0.4875	{max_depth: 1}, {min_samples_leaf: 1}, {min_samples_split: 2}

Decision tree's optimal selection maximum depth of one corresponds to a single decision rule, indicating a strong preference for an extremely simple model structure. Increasing tree depth beyond this point led to overfitting without improving generalization. This result suggests that default risk is largely driven by one dominant feature, with additional splits primarily capturing noise.

While the decision tree achieved a higher validation score than logistic regression, this advantage did not translate to improved test performance, highlighting the stabilizing effect of logistic regression's strong regularization.

3. K-nearest neighbors

Val	Test	Params
0.3118	0.2768	{n_neighbors: 1}, {weighting: uniform}

kNN performed substantially worse than all other evaluated models. The optimal selection of $k = 1$ indicates that the local neighborhood structure is noisy and unstable, as aggregating information across multiple neighbors only served to degrade performance.

The poor performance of kNN is to be expected, as the algorithm suffers from the curse of dimensionality and lacks a mechanism for regularization and handling class imbalance.

B. Stochastic Models

4. Random Forest

Seed	Val	Test	Params: max_depth, max_features, n_estimators		
42	0.5824	0.4851	{2},	{0.2},	{200}
84	0.5851	0.4756	{1},	{0.3},	{200}
126	0.5878	0.4752	{1},	{0.1},	{100}

Across all seeds, the model consistently favored very shallow trees and aggressive feature subsampling, reinforcing the decision tree results and suggesting that default risk is driven by a small number of dominant features. Restricting tree depth and limiting the feature set at each split appears to reduce overfitting from chance correlations.

The number of estimators varied between 100 and 200 across seeds, with no consistent performance advantage from additional trees. This indicates that model variance was already sufficiently reduced at relatively low ensemble sizes, and that adding more trees provided diminishing returns.

Notably, random forests did not improve upon the decision tree baseline and, on average, performed slightly worse on the test set. This suggests that ensembling offers limited benefit when the optimal base learner is already extremely simple, instead possibly diluting the dominant signal rather than improving generalization.

5. XGBoost

Seed	Val	Test	Params: max_depth, reg_alpha, reg_lambda		
42	0.5830	0.4944	{1},	{20},	{1500}
84	0.5838	0.4932	{1},	{1},	{2000}
126	0.5840	0.4950	{1},	{20},	{1000}

Across all seeds, the model opted for extremely shallow trees with a max depth of 1, as well as very high l2 regularization (1000 - 2000) and moderate to high l1 regularization (1-20). This

configuration corresponds to boosting heavily regularized decision stumps, allowing the model to make small sequential corrections while avoiding overfitting.

Across all algorithms, models consistently favored extreme simplicity and strong regularization, suggesting that default risk is governed by a small number of dominant features. The strong performance of logistic regression implies that much of the predictive structure is monotonic, as increased nonlinear capacity offered limited gains beyond those achieved by XGBoost.

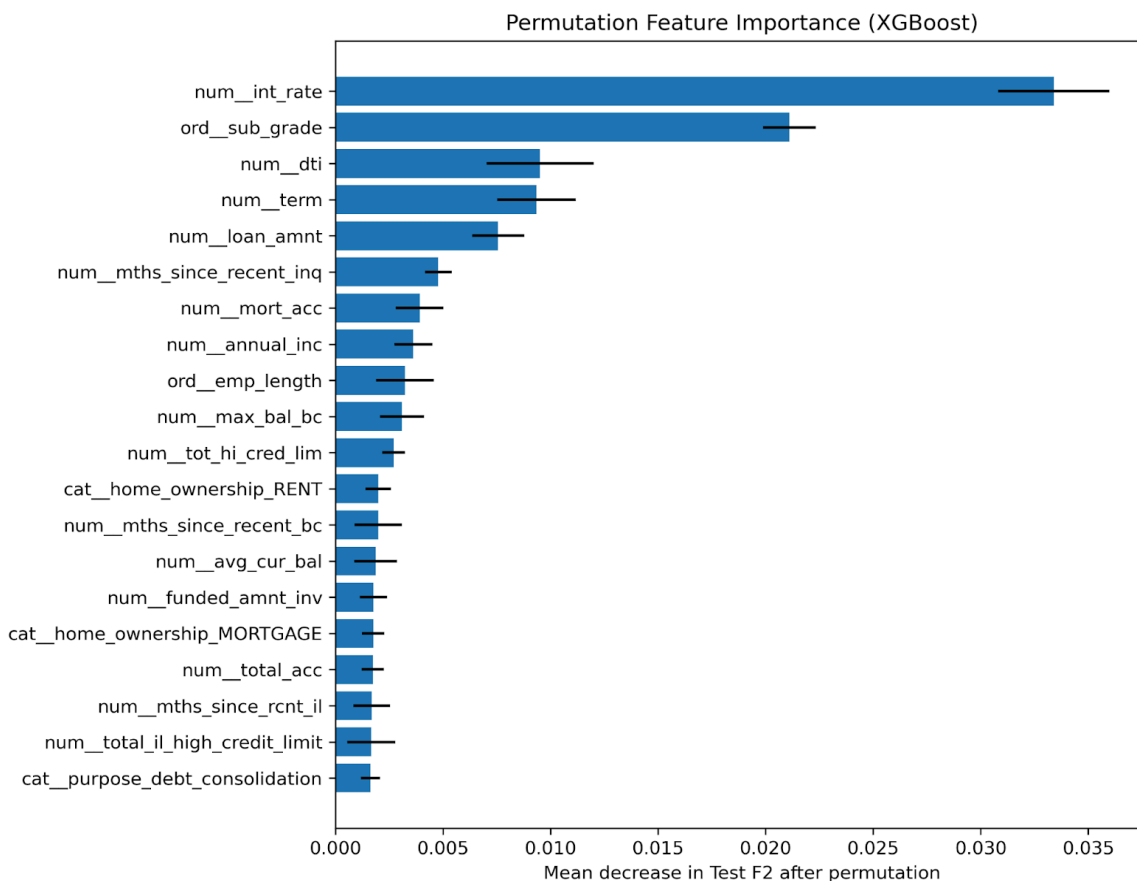
As XGBoost achieved the strongest predictive performance, all subsequent global and local feature importance analyses are conducted using this model.

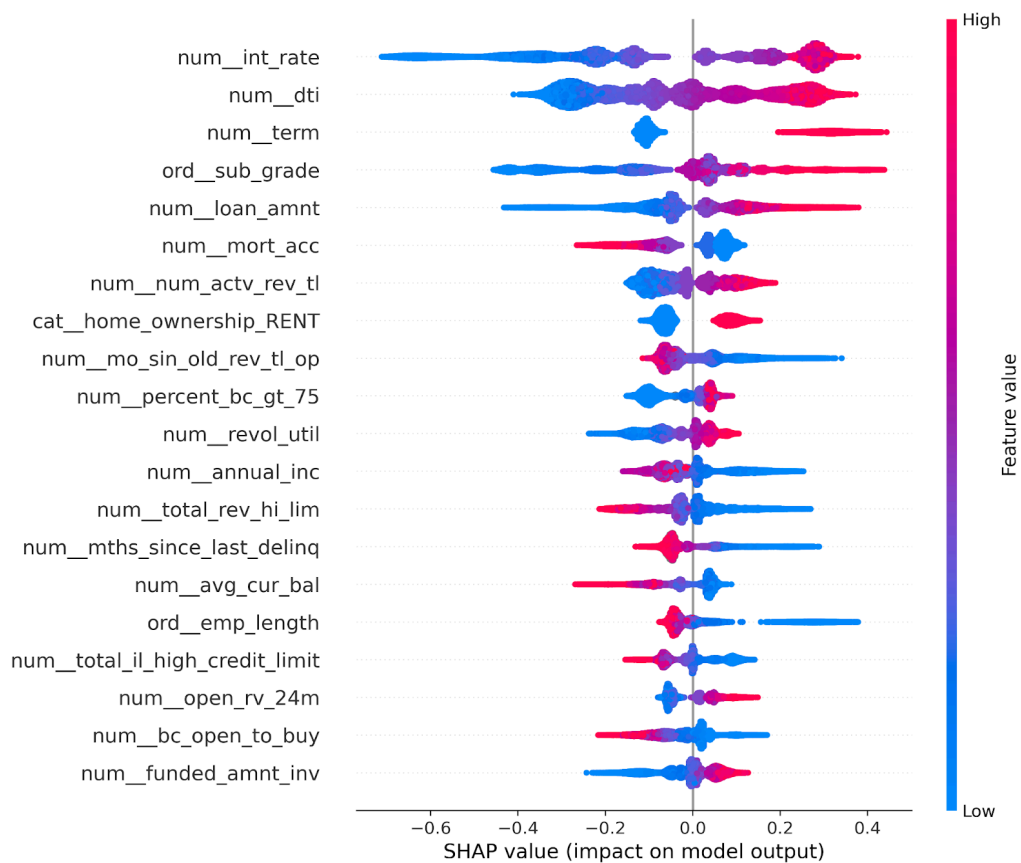
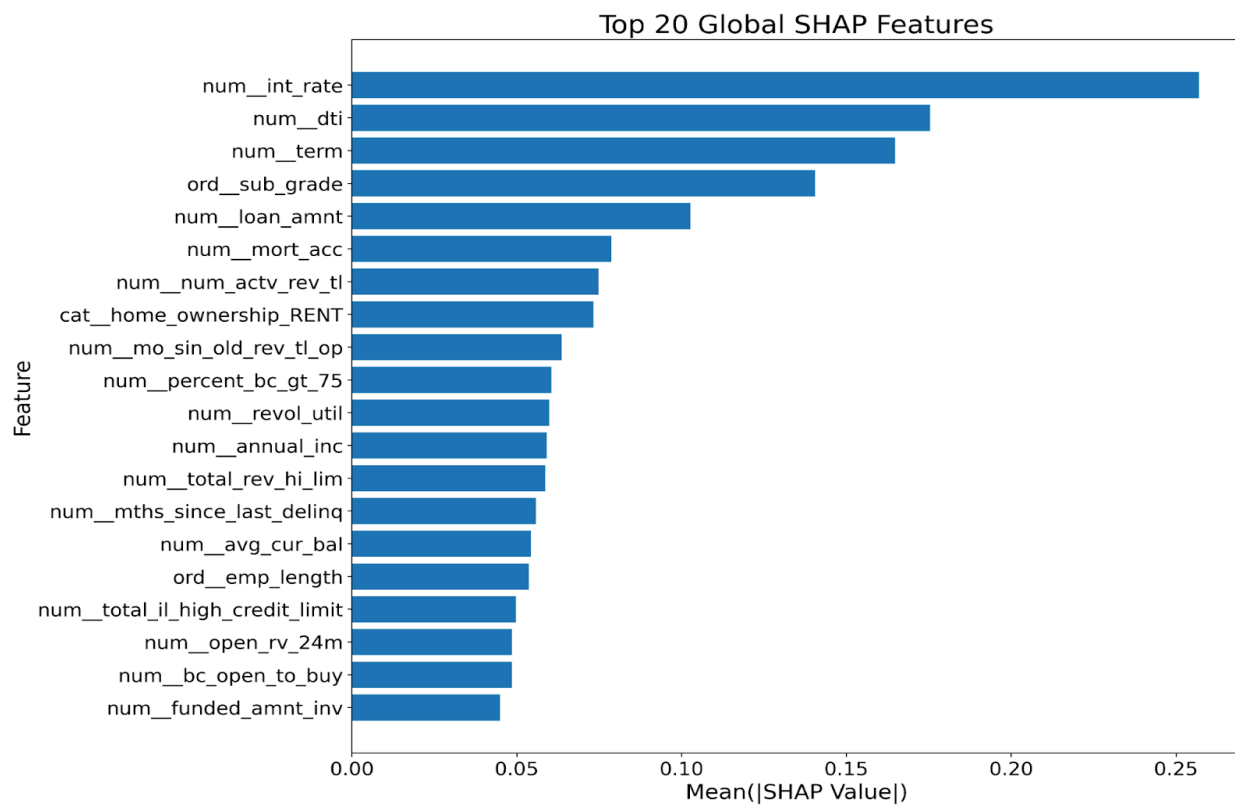
2. Global Importance

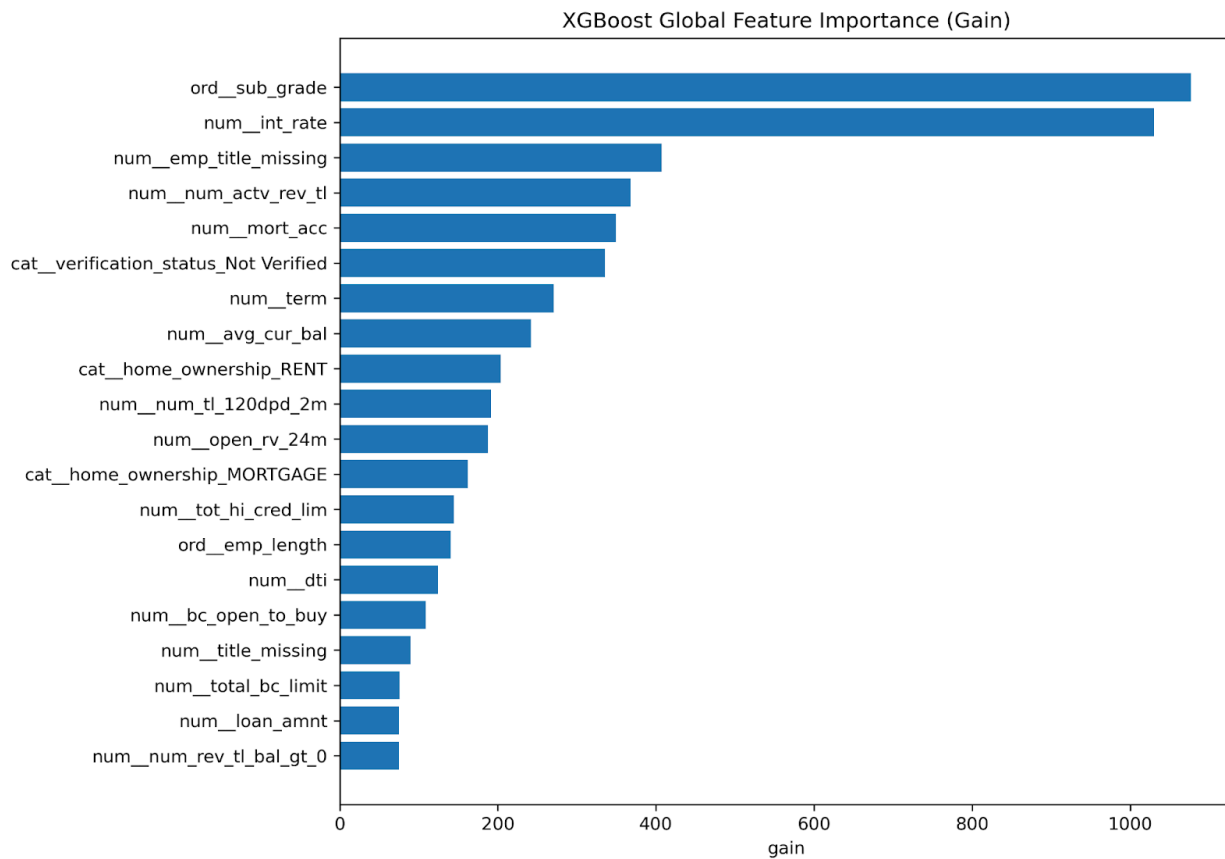
Global feature importance was analyzed using four complementary approaches: permutation feature importance (Figure 5), mean absolute SHAP values (Figure 6), SHAP beeswarm plots (Figure 7), and XGBoost split gain (Figure 8). All methods identify a consistent set of dominant predictors, most notably interest rate, followed by measures of borrower debt burden and loan structure such as debt-to-income ratio, loan sub-grade, loan term, and loan amount.

The SHAP beeswarm plot further reveals that the relationships between these top features and default risk are largely monotonic, consistent with the strong performance of linear models such as logistic regression. The strong agreement across all importance measures suggests that model predictions are driven by a small number of signals rather than complex nonlinear interactions.

Figure 5 - 8:



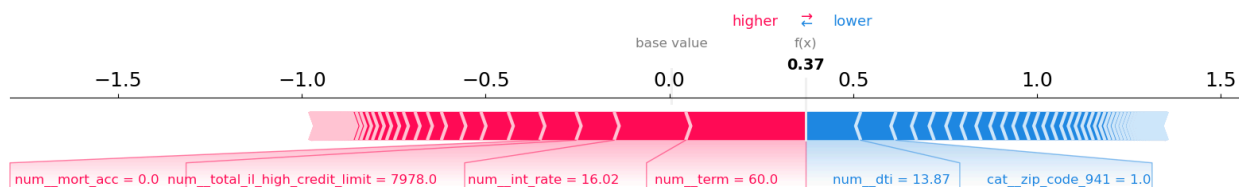




3. Local Importance

Local feature importance provides insight into how individual features contribute to a prediction for a specific observation. The horizontal axis on the below force plots represent log odds, where the base value (~ 0.005) corresponds to a 50% predicted probability of default. Positive SHAP values increase default risk, while negative values decrease it.

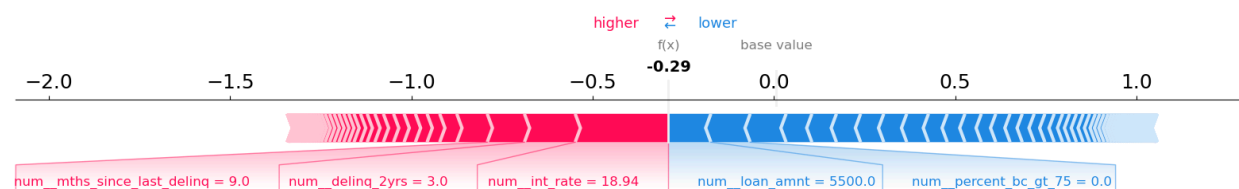
Figure 9



This first observation has a log-odds value of 0.37, corresponding to a 59% probability of default. The factors increasing default risk are a longer loan term, moderately high interest rate, and indicators of limited credit history, including a low total credit limit and zero mortgage accounts.

Factors reducing default probability include a relatively low debt to income ratio, and the borrower's zip code (San Francisco). Geographic indicators may capture correlated regional risk characteristics and should be interpreted carefully.

Figure 10:



The second observation has a log-odds value of -0.29, corresponding to a 43% probability of default. Features increasing default risk include a high interest rate and evidence of recent delinquency, such a recent delinquency 9 months ago and 3 delinquencies in the past 2 years.

In contrast, features reducing default risk include a relatively small loan amount and indicators of safe credit behavior, such as no bank cards above 75% utilization.

Outlook

Several avenues could be explored to further improve both predictive accuracy and interpretability. First, across all models, a wider range of hyperparameters could be tuned to better ensure that optimal configurations are reached. Additionally, increasing the number of folds in the time-series split and evaluating more random seeds would provide more reliable estimates of model uncertainty and stability.

Further gains may come from feature engineering, particularly through the inclusion of interaction terms and variables that capture temporal drift. For example, incorporating loan origination quarters and macroeconomic indicators (CPI, unemployment) could help account for changes in lending conditions over time.

Lastly, explicit threshold tuning on a separate validation slice could also improve model performance, though this would complicate cross-model comparison by introducing model-specific thresholds.

From an interpretability perspective, local explanations could be improved by fixing SHAP axis scales across observations and exploring multi-feature perturbation. Finally, uncertainty arising

from iterative imputation could be quantified by repeating imputations across multiple random seeds.

References

Dataset: <https://www.kaggle.com/datasets/adarshsng/lending-club-loan-data-csv/data>

Previous work:

<https://www.kaggle.com/code/beatafaron/credit-risk-eda-defaults-segments-trends-1>

<https://www.kaggle.com/code/faressayah/lending-club-loan-defaulters-prediction>